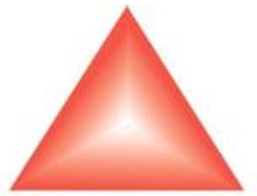


جامعة محمد الخامس بالرباط
Université Mohammed V de Rabat

ROYAUME DU MAROC
MINISTÈRE DE L'ENSEIGNEMENT
SUPÉRIEUR, DE LA RECHERCHE
SCIENTIFIQUE ET DE L'INNOVATION



المملكة المغربية
وزارة التعليم العالي
و البحث العلمي و الإبتكار



المدرسة الوطنية العليا للمعلوماتية و تحليل النظم
École Nationale Supérieure d'Informatique
et d'Analyse des Systèmes

Doctoral Studies Center in Information and Engineering Sciences and Technologies (ST2I)

Cooperative Game Theory for Explainable AI in Recommendation Systems: A Shapley Framework for Actionable Insight

PhD Viva Presented by: Mouad LOUHICHI

Jury members:

President	[President to complete]		[Institution]
Supervisor	Pr. Mohamed LAZAAR	PES	ENSIAS, Mohammed V University, Rabat
Reviewer	[Reviewer to complete]		[Institution]
Reviewer	[Reviewer to complete]		[Institution]
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Motivation: Three Questions

Motivation

Actionable Insight

Ubiquity

How do opaque AI systems shape what billions of users see, buy and watch every day?

The Black Box

Why do state-of-the-art recommenders and clustering pipelines remain opaque to users and designers?

Toward Trust

How can transparency be built as part of the model, instead of being bolted on afterwards?

tension: as models gain expressive power, they lose the transparency needed for trustworthy deployment

Ubiquity

Recommenders shape news, study, health, credit

Market > \$15B by 2029

Black box

Deep & graph models: opaque, non-auditable

EU AI Act: high-risk -> explain

Toward trust

Accountable, auditable, modifiable insight

Actionable, not just accurate

▪ Definition 1.1 (Actionable insight)

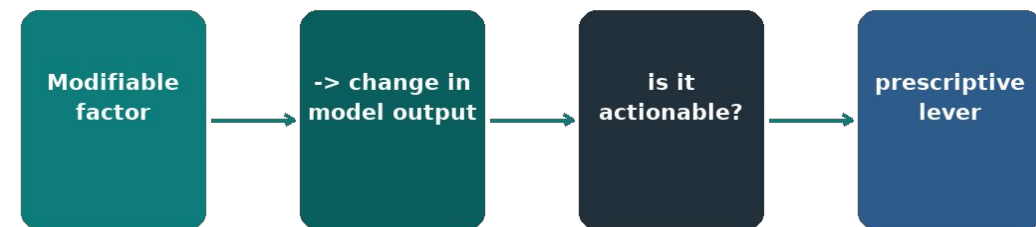
- An explanation is actionable when it identifies at least one modifiable factor whose change is associated with a specifiable change in model output...
- ... and that factor is expressible in the semantic vocabulary of the task domain.

▪ Accessibility means the domain's own terms:

- a physicochemical variable (wine), a pollution indicator (air quality), or a preference signal (recommendation).
- not an opaque latent code.

▪ Why: an explanation that identifies a modifiable driver supports intervention, not merely description.

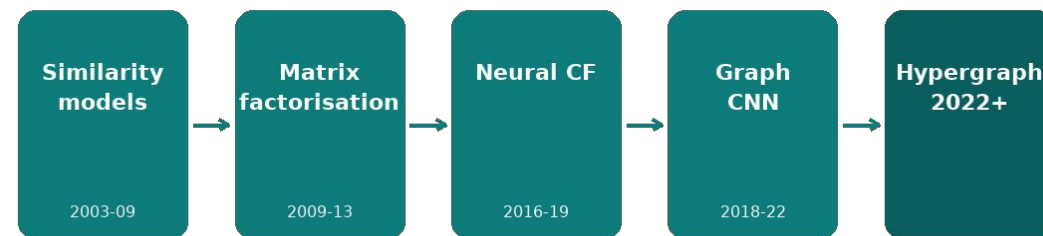
From explanation to action



Actionable = a real-world lever you can pull to change the outcome

- Recommenders evolved from similarity filters to complex representation-learning systems on sparse, high-dimensional, dynamic data.
 - MF → neural CF → graph CNNs → hypergraph: each step improved ranking but intensified the interpretability deficit.
- Why the deficit matters:
 - Undermines user trust.
 - Constrains debugging and scientific learning.
 - Collides with regulatory expectations (EU AI Act, OECD principles, GDPR).
 - EU AI Act (Art. 13) requires high-risk systems to provide explanations in human-understandable terms.

Evolution of recommender systems



Interpretability deficit grows: opaque embeddings, non-modifiable factors

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AI-Powered Recommendation Systems Around Us

Motivation

Actionable Insight

Research Context

- **AI recommendation mediates what billions see, buy and listen to every day:**

- Streaming & video: Netflix, Prime Video, YouTube.
- Music: Spotify, YouTube Music, Deezer.
- Shopping & e-commerce: Amazon, AliExpress, Noon.
- Social & feeds: TikTok, Instagram, LinkedIn.
- Maps, ads, search: Google Maps, Uber, Google Ads, Bing.

- **Accurate yet opaque — the very gap our thesis targets.**

AI-Powered Recommendation Is Everywhere

Streaming & Video

Netflix, Prime Video, YouTube

Music

Spotify, YouTube Music, Deezer

Shopping & E-commerce

Amazon, AliExpress, Noon

Social & Feeds

TikTok, Instagram, LinkedIn

Maps & Local

Google Maps, Uber, Booking

Ads & Search

Google Ads, Bing, Meta

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Context & Problematic

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Recommendation & Clustering Paradigms

Paradigms

Problematic

Contributions

- Collaborative filtering – users who behaved similarly will value similar items (user-/item-based).
- Content-based – recommends items sharing attributes with a user profile.
- Hybrid – combines collaborative and content signals.
- Matrix factorisation – $R \approx PQ^T$: compact but opaque latent factors.
- Graph-based – interaction graph with neighbourhood propagation (LightGCN, hypergraph).

Content-Based Filtering (explainable example)



Example: liked sci-fi + Nolan -> surface films with similar genre / cast / keywords

Candidate item features

genre = Sci-Fi director = Nolan
cast = ... year = 2010 rating = 8.6

Why recommended (explainable)

high cosine similarity to your profile
sim = cos(f_user, f_item)

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Paradigms

Problematic

Contributions

- Data sparsity & scalability – the user–item matrix is overwhelmingly empty.
- Cold-start – new users and items are structurally disadvantaged.
- Popularity bias & lack of diversity – exposure begets interaction, begets exposure.
- **Absence of interpretability – the most fundamental limit.**
 - For clustering: methods favour local OR global explanation, not both.
 - They struggle to scale, and explanations rarely stay coherent across resolutions.

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Three Structuring Limitations (Problem Statement)

Paradigms

Problematic

Contributions

- **1. Lack of explainability** – complex models remain hard to interpret faithfully and actionably.
- **2. Difficulty of scaling** – local explanations do not transfer naturally to hierarchical structures or large datasets.
- **3. Weak integration into learning** – most explanations stay post-hoc and do not shape model dynamics or the accuracy-diversity-context trade-off.
 - Thesis gap: the literature still lacks a single cooperative-attribution framework that explains clustering faithfully, stays coherent under hierarchy, and then operates as an in-training signal in recommendation.

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Research Questions (RQ1–RQ5) and Overall Aim

Paradigms

Problematic

Contributions

- **Aim: develop, justify and evaluate a cooperative-game-theoretic perspective for XAI in clustering and recommendation.**
- RQ1 · How can Shapley values explain black-box clustering faithfully at instance and cluster level?
- RQ2 · How can this extend to large-scale, hierarchical clustering without losing tractability or consistency?
- RQ3 · Can cooperative attribution move beyond post-hoc and enter the learning dynamics of graph recommenders?
- RQ4 · Can a recommender jointly optimise ranking accuracy, context and diversity when importance is estimated by a cooperative-game utility?
- RQ5 · What emerges when clustering explanation and recommendation learning are two stages of one cooperative-game perspective?

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The Three Contributions

Paradigms

Problematic

Contributions

- **C1 – Explainable black-box clustering: PCA–K-Means–LightGBM–TreeSHAP pipeline.**
 - → Wine Quality: faithful instance- and cluster-level feature attribution.
- **C2 – Enhanced multi-level XAI for large-scale clustering with cross-level SHAP aggregation.**
 - → Beijing Air Quality: hierarchical attribution consistency under scale.
- **C3 – DyHuCoG: Dynamic Hypergraph Cooperative Game for preference-aware recommendation.**
 - → MovieLens-1M & Amazon-Book: preference-aware Monte Carlo Shapley as an in-training signal.
 - **Thesis claim: cooperative game theory functions as a shared attribution perspective for explanation, optimisation and intervention.**

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Paradigms

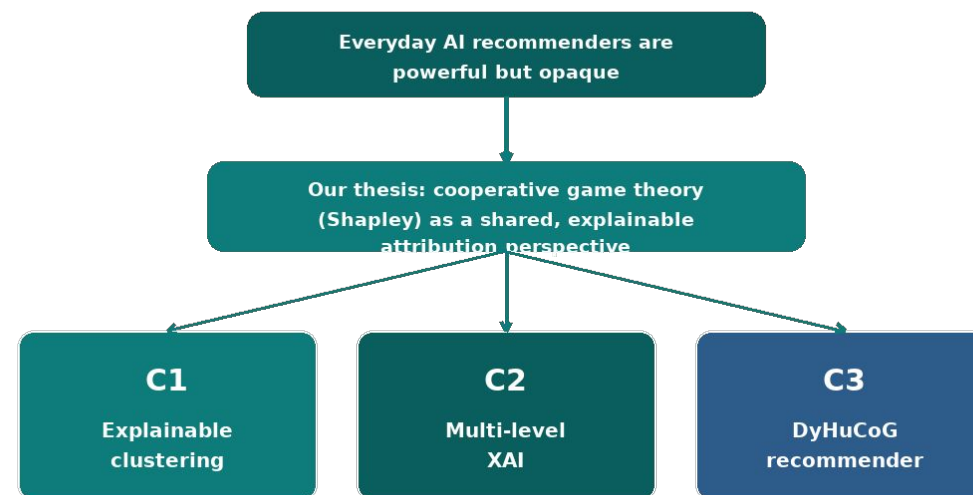
Our Thesis

Problematic

Contributions

- **Thesis: cooperative game theory (Shapley) is one shared, explainable attribution perspective.**
- The same logic spans clustering and recommendation:
 - C1 – explains black-box clustering faithfully (wine).
 - C2 – keeps explanation coherent under scale and hierarchy (Beijing).
 - C3 – DyHuCoG turns attribution into an in-training signal.
- **From post-hoc description to in-training guidance.**

Our Thesis in One View





Experimental Protocol

Dataset	Scale	Type	Role
Wine Quality (vinho verde)	4,898 × 11	Tabular, numeric	C1 – single-level clustering
Beijing Multi-Site Air Quality	383,585 × 11	Tabular, pollutant + meteorology	C2 – multi-level clustering
MovieLens-1M	6,040 u / 3,706 i / 1.0M int	Implicit feedback (0.0447)	C3 – DyHuCoG
Amazon-Book	52,643 u / 91,599 i / 3.0M int	Implicit feedback (0.0006)	C3 – DyHuCoG

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Dataset Characteristics & Why Each Was Chosen

Datasets

Metrics

Hardware

- Wine Quality – 11 interpretable physicochemical features; small, dense, chemically correlated → tests attribution in a trusted semantic space.
- Beijing Air Quality – 11 pollutant + meteorological variables, 383,585 hourly records → tests scale, noise and temporal/meteorological structure.
- MovieLens-1M – ~1.0M interactions, density 0.0447 → benchmark-standard recommendation with established baselines.
- Amazon-Book – ~3.0M interactions, density 0.0006 → deliberately sparse to stress-test Shapley weighting when signal is weak.
- **The 0.0447 vs 0.0006 density contrast is intentional: the claim is that gains are largest where data are weakest.**

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Data Splitting & Preprocessing

Datasets

Metrics

Hardware

- Clustering: five-fold cross-validation for surrogate/attribution stability.
- Recommendation: user-level, temporal split — 70% train / 10% val / 20% test.
- Leave-one-out: the latest test positive per user is the target, ranked against negatives.
- Implicit conversion: MovieLens-1M ratings > 3 treated as positive.
- Popularity-aware negative sampling: $q(i) \propto f_i^\eta$ for harder contrasts.
- Reproducibility: seeds {42,43,44,45,46}; early-stopping patience 20.

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- **Clustering benchmark: LIME-based surrogate explanation pipeline.**
- **Recommendation benchmarks: MF, NCF, LightGCN, RecDCL, HCCF, HPCF (strongest reference).**
- Ranking: Precision@K, Recall@K, NDCG@20 (principal).
- System diversity: Catalogue Coverage = $|\bigcup R_u| / |I|$.
- List diversity: Intra-List Diversity (ILD), built into the coalition utility.
- Clustering quality: Silhouette, Davies–Bouldin.

Evaluation metrics

Ranking

- NDCG@k
- Recall@k
- MAP@k
- HR@k

Beyond-accuracy

- Coverage
- ILD
- Novelty
- Fairness

Clustering validity

- Silhouette
- Davies-Bouldin
- Calinski-H
- Stability

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Hardware & Software

Datasets

Metrics

Hardware

- CPU: Intel Core i9-14900K, 24 cores — clustering, preprocessing, data loading.
- GPU: NVIDIA GeForce RTX 4090, 24 GB — DyHuCoG training & inference.
- RAM: 48 GB; Storage: 2 TB SSD.
- Python 3.8; scikit-learn, LightGBM, SHAP, PyTorch 2.0.1, NumPy/SciPy/pandas.
- Altair for interactive SHAP visualisation; metrics at $K \in \{5, 10, 20\}$.

4

Contribution I — Explainable Black-Box Clustering

Objectives

Methodology

Results

Findings

- **Gap: Shapley explanation is dominant in supervised tasks, but clustering remains comparatively under-explained.**
 - Existing clustering-interpretability methods privilege local or global explanation, not both.
 - They often fail to scale or preserve coherence across clusters.
- **Objectives: build a pipeline yielding cluster-level explanation while preserving feature-level attribution.**
 - Preserve the semantics of the original feature space; justify why Shapley is better than LIME.

- **RQ1 · How can Shapley values explain black-box clustering faithfully at instance and cluster level?**
 - Objective O1 · Build a pipeline that yields cluster-level explanation while keeping feature-level attribution.
 - Objective O2 · Preserve the semantics of the original feature space, not a reduced latent space.
 - Objective O3 · Justify Shapley over an ad-hoc surrogate such as LIME.

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Objectives

Methodology

Results

Findings

- **Player set $N = F$ — each feature is a player.**
- Value function $v(S) = \text{Silhouette}(\text{KMeans}(X_S, k^*))$ — how well data cluster using only features in S .
- A feature's Shapley value = its expected marginal contribution to clustering quality over all coalition orders.
- Why Silhouette: bounded, normalised, semantically intuitive.
- **Direct evaluation for every coalition is intractable → we need a bridge.**

- **Once K-Means produces cluster labels, train a LightGBM multiclass surrogate to predict them from original features.**
- Apply TreeSHAP to the surrogate — fast, exact tree-based attribution in the original semantic feature space.
 - Direct TreeSHAP on K-Means is impossible (it explains tree models, not centroids).
 - The surrogate preserves the chemistry/pollution vocabulary that makes the analysis actionable.
 - **Validity condition: surrogate fidelity is high (macro-F1 $\approx$ 0.82).**

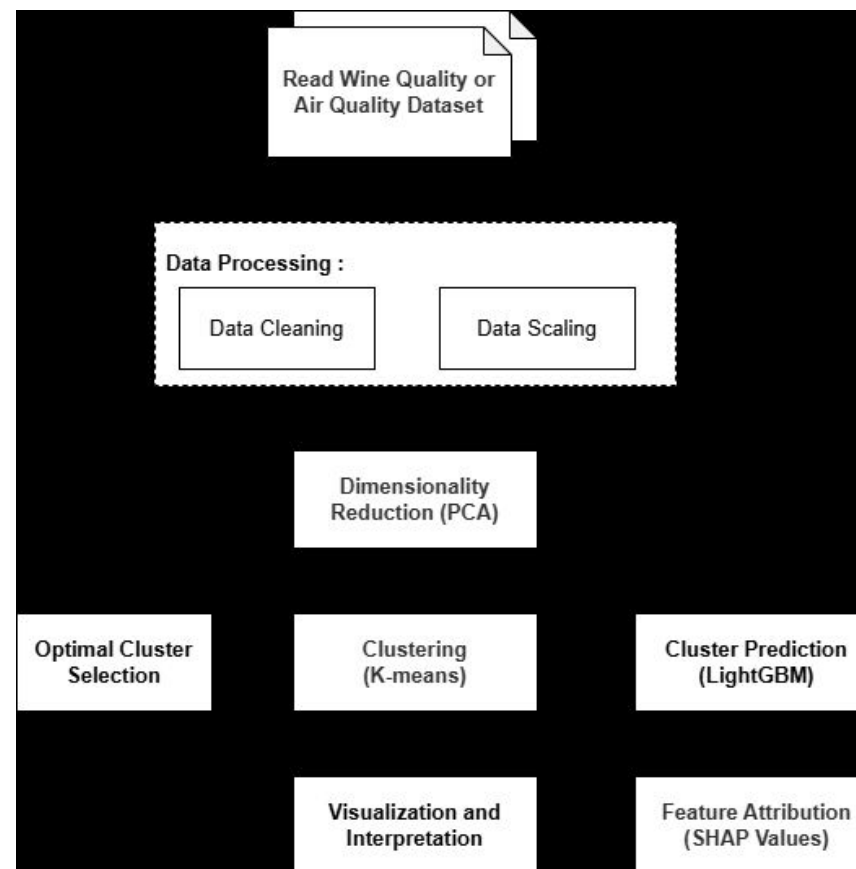
Objectives

Methodology

Results

Findings

- Stage 1 – PCA: stabilise geometry + visual diagnostic. NOT the explanatory space.
- Stage 2 – K-Means++ with multi-criteria k selection (elbow, Silhouette, Davies–Bouldin).
- Stage 3 – LightGBM surrogate trained on original features to predict cluster labels.
- Stage 4 – TreeSHAP attribution in the original feature space.
- Stage 5 – Aggregate into global importance, cluster-specific profiles, and local force plots.
 - Complexity: dominated by PCA and repeated K-Means; TreeSHAP scales with tree count, not exponentially in features.



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Objectives

Methodology

Results

Findings

- Multi-criteria evaluation across $k \in \{2..10\}$ using elbow, Silhouette, Davies–Bouldin.
- **We select $k^* = 3$ — even though it is NOT geometrically optimal:**
 - $k = 2$: Silhouette 0.214, Davies–Bouldin 1.775 (better separation).
 - $k = 3$: Silhouette 0.144, Davies–Bouldin 2.097 (weaker separation).
 - **Why: three clusters give a semantically richer oenological partition → more actionable.**
 - Note: the higher Silhouette ≈ 0.63 belongs to Beijing (C2), not this wine partition.

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Objectives

Methodology

Results

Findings

- **Global SHAP ranking (high → low):**

- density → pH → fixed acidity → sulfur-dioxide → alcohol
- Dominant drivers relate to structure, preservation, and sensory balance.
- **This is NOT an arbitrary classifier artefact — it recovers a chemically interpretable hierarchy.**

- **Three clusters show distinct explanatory signatures.**

- Cluster 0 – density + sulfur-dioxide-related variables.
- Cluster 1 – acidity and pH-related effects.
- Cluster 2 – a different balance of acidity, alcohol, and related chemical attributes.
- The same small set of variables recurs across clusters, with different relative weights within each.

Objectives

Methodology

Results

Findings

Criterion	SHAP (cooperative)	LIME (local surrogate)
Basis	Cooperative-game marginal contribution	Local surrogate approximation
Local / global	Both	Primarily local
Theoretical guarantee	Efficiency, symmetry, null player, additivity	None equivalent
Stability	Higher when surrogate faithful	Sensitive to perturbation design
Cluster comparison	Strong	Limited

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Objectives

Methodology

Results

Findings

- **Answers to the questions and objectives**

- RQ1 answered in the affirmative — faithful, coherent cluster-level explanation from Shapley values.
- O1 met · cluster-level explanation while preserving feature-level attribution.
- O2 met · attribution returned to the original chemical variables.
- O3 met · Shapley grounded in four axioms; LIME has no equivalent guarantee.

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Objectives

Methodology

Results

Findings

▪ Key findings

- Cluster-level explanation anchored to individual feature contributions.
- Explanations returned to the original chemical variables, not a latent space.
- Theoretically grounded case for Shapley over LIME (efficiency, symmetry, null player, additivity).
- Recovers an oenologically interpretable ranking — density, pH, acidity, sulfur dioxide, alcohol.

▪ Limitations

- Fidelity depends on the LightGBM surrogate — not a direct mechanism of the K-Means geometry.
- Confined to tabular data; no structured, graph, or image input.
- Single-level structure: cannot explain how importance reconfigures between a partition and its sub-partitions.
- The surrogate approximation compresses observation-level variation.

- **Takeaways**

- Shapley attribution is a single, principled lens for explaining an unsupervised partition.
- Keeping attribution in the original feature space is what makes it actionable.
- **But real data are rarely single-level: broad regimes contain nested sub-groups.**

- **So the next question is whether this logic survives scale and hierarchy.**

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Contribution II — Enhanced Multi-Level XAI for Large-Scale Clustering

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Objectives

Methodology

Results

Findings

- **Gap: once clustering is multi-level, feature importance must stay interpretable within a cluster, across sub-clusters, and across the hierarchy.**
 - Large-scale data make exact explanation computationally burdensome.
 - Flat explanation may be true yet incomplete — it cannot show how importance changes inside a cluster.
- **Objectives: a genuinely multi-level workflow; a formal cross-level consistency argument (Prop. 6.1); validation on a structurally different large-scale dataset.**

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Objectives

Methodology

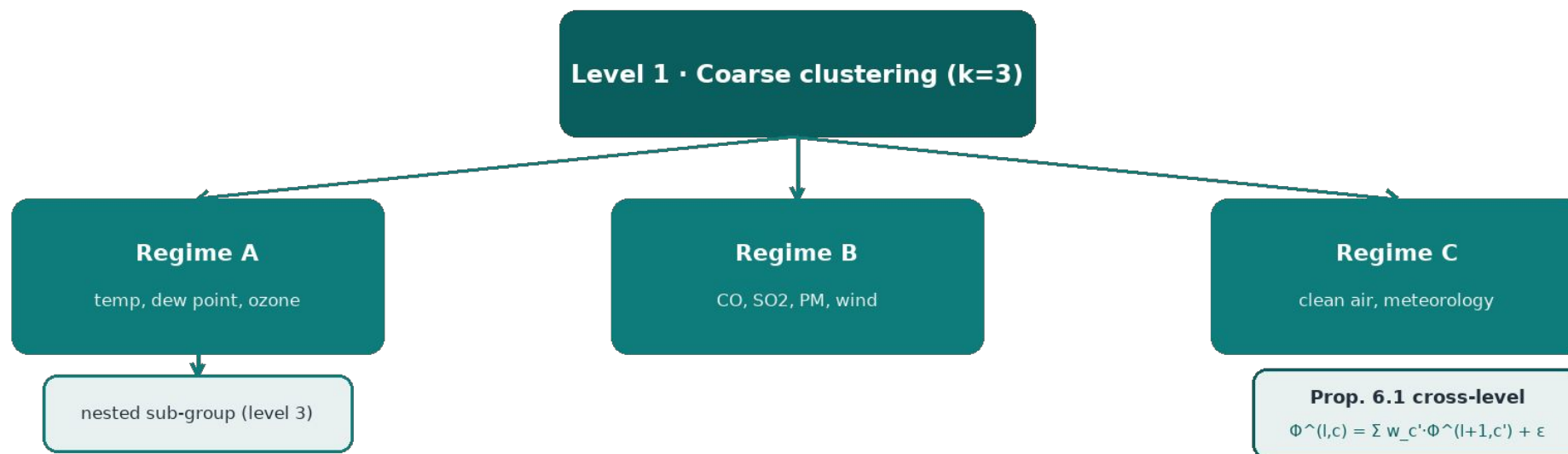
Results

Findings

▪ **RQ2 · How can this extend to large-scale, hierarchical clustering without losing tractability or consistency?**

- Objective O1 · A genuinely multi-level workflow, not a rerun of the single-level pipeline.
- Objective O2 · A formal cross-level consistency argument (Proposition 6.1).
- Objective O3 · Validation on a structurally different large-scale dataset.

- **Recursive/nested: coarse clustering on the full dataset, then subdivide each cluster.**
- For each level, train a level-specific surrogate and compute SHAP in the SAME original feature space.
- **Cross-level aggregation is NOT a naive average — it respects cluster size and nesting structure.**
 - Parent-level attribution = an expectation over the explanatory structure of its descendants.
 - The hierarchy is a pragmatic analytical device, not a claim of true ontological hierarchy.



- Let $\Phi^{\wedge(l,c)}_j = E_{\{x \sim c\}}[|\phi_j^{\wedge(l)}(x)|]$ be the expected absolute SHAP importance of feature j at level l in cluster c .
- Let $w_{c'} = |c'| / |c|$ be the relative size of child c' within parent c .
- **For a strict nested hierarchy on a consistent feature space:**
 - $\Phi^{\wedge(l,c)}_j = \sum_{\{c' \in \text{child}(c)\}} w_{c'} \cdot \Phi^{\wedge(l+1,c')}_j + \epsilon_j$
 - ϵ_j is a residual from surrogate mismatch, vanishing under perfect fidelity.
 - Derived via law of total expectation (children partition the parent).
 - Does NOT imply explanations are identical across levels — it implies differences can be interpreted, not dismissed as inconsistency.

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Objectives

Methodology

Results

Findings

- **Full dataset, $k = 3$ (strong convergence on multi-criteria evaluation).**

- Silhouette ≈ 0.63 — materially stronger separation than wine.
- Davies–Bouldin ≈ 0.55 — low between-cluster ambiguity.
- PCA projection (2 components) used only for visual inspection.
- Sensitivity: robust to modest variation in k , projection dim, surrogate depth; only low-ranked variables shift.

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Objectives

Methodology

Results

Findings

- **Global SHAP ranking (high → low):**

- temperature → dew point → pressure → CO → NO₂ → PM₁₀ → PM_{2.5}
- It is NOT simply pollutant concentrations that matter — meteorological variables play a structurally central role.
- Temperature, dew point and pressure condition dispersion, trapping, and photochemical behaviour.
- This is the kind of insight flat descriptive summaries often fail to make explicit.

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Objectives

Methodology

Results

Findings

- Regime A – warm photochemical: ozone, temperature, dew point prominent (summer photochemical smog).
- Regime B – wintertime smog: CO, SO₂, PM dominate; low wind speed suppresses dispersion.
- Regime C – comparatively clean air: favourable meteorology, weak pollutant pushes.
 - **The framework shows not only that these regimes exist, but which variable combinations define them.**

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Objectives

Methodology

Results

Findings

- At the coarse level, temperature and dew point dominate — they differentiate broad atmospheric regimes.
- Within individual clusters, CO, SO₂, PM₁₀, wind speed, pressure, or ozone become more discriminative.
- **This change is NOT contradictory — it is exactly what a multi-level explanation should reveal.**
 - Parent-level story = regime selection. Cluster-level story = variation within a regime.
 - A variable can be globally important yet locally uninformative within a sub-cluster.

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Objectives

Methodology

Results

Findings

- Wine: small, dense, chemically correlated. Beijing: large, noisy, temporally and meteorologically variable.
- **The same explanatory logic remains productive in both → not tied to one domain-specific peculiarity.**
 - vs SHAP-based clustering literature: Beijing Silhouette ≈ 0.63 vs Gramegna & Giudici credit-risk 0.37.
 - LIME comparator: weaker structural coherence, less stable local narratives for hierarchical reasoning.

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Objectives

Methodology

Results

Findings

- **Answers to the questions and objectives**

- RQ2 answered in the affirmative, with bounds — coherence retained under scale and hierarchy.
- O1 met · a multi-granular explanation that does not collapse into a single flat summary.
- O2 met · Proposition 6.1 provides a formal cross-level consistency argument.
- O3 met · validated on a structurally different large-scale dataset (Beijing air quality).

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Objectives

Methodology

Results

Findings

▪ Key findings

- Scalable, multi-granular explanation that does not collapse into a single flat summary.
- Formal cross-level consistency argument (Proposition 6.1).
- Validated on a structurally different large-scale dataset (Beijing air quality).
- Differences across levels are interpretable, not dismissed as inconsistency.

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Objectives

Methodology

Results

Findings

▪ Limitations

- Clustering remains static, even though the Beijing data are temporal.
- Surrogate-based SHAP plus representative-instance reporting compress observation-level variation.
- Confined to tabular data.
- **Still an explanation of a pre-computed partition — it does not influence learning.**

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Objectives

Methodology

Results

Findings

- **Takeaways**

- Shapley attribution stays coherent across granularity — when the hierarchy is explicit.
- Explanations become interpretable against scale, not just against a single flat partition.
- **But the attribution is still post-hoc: it explains a partition that was already computed.**

- **So the next step is to let attribution shape the learning itself.**



Contribution III — DyHuCoG: A Dynamic Hypergraph Cooperative Game

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Objectives

Methodology

Results

Findings

- **Gap: graph and hypergraph recommenders treat message importance as either uniform or attention-weighted, without a principled marginal-contribution account.**
 - Diversity is often a secondary objective or a re-ranking heuristic.
 - Interpretability is added after prediction, not integrated into the learning objective.
- **Objectives: formulate recommendation as a cooperative game; embed preference-aware Monte Carlo Shapley into message passing; improve ranking, coverage, and diversity jointly.**

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Objectives

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Results

Findings

▪ **RQ3 · Can cooperative attribution move beyond post-hoc and enter the learning dynamics of graph recommenders?**

- RQ4 · Can a recommender jointly optimise ranking accuracy, context and diversity when importance is estimated by a cooperative-game utility?
- Objective O1 · Formulate recommendation as a cooperative game with users, items and contexts as players.
- Objective O2 · Embed preference-aware Monte Carlo Shapley into hypergraph message passing.
- Objective O3 · Improve ranking, coverage and diversity jointly.

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Objectives

Methodology

Results

Findings

- **Player set $N = U \cup I \cup C$ (users, items, contexts).**
- Hypergraph $H = (V, E, W)$; $V = U \cup I \cup C$; W = dynamic edge weights from Shapley estimates.
- Coalition $S \subseteq N$ represents entities participating in a recommendation episode.
- Coalition value $v(S)$ measures the quality of the recommendation outcome achievable by S .
- Top-N task: produce a ranked list L_u balancing relevance, diversity, and contextual fit.

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Objectives

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Findings

- **$v(S) = \alpha \cdot \text{NDCG@20}(S) + \beta \cdot \text{Diversity}(S) + \gamma \cdot \text{ContextScore}(S)$, with $\alpha + \beta + \gamma = 1$.**
 - The same trade-off the recommender must satisfy is the trade-off from which attribution is computed — explanatory game and predictive objective are aligned by design.
- **Preference-weighted: $v_{\text{pref}}(S) = v(S) + \lambda_{\text{pref}} \cdot \sum_{\{(u,i) \in S\}} \text{sim}(u,i)$.**
 - $\alpha = 0.60, \beta = 0.25, \gamma = 0.15; \lambda_{\text{pref}} = 0.20$ — grid-searched, stable (<1.5% variance in NDCG@20).
 - Coalition evaluation scoped to the interaction episode (a few dozen players), not the full catalogue.

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Objectives

Methodology

Results

Findings

- Exact Shapley is combinatorial and infeasible for realistic systems.
- **Monte Carlo estimator:** $\hat{\phi}_j = (1/M) \sum_m [v(S_m \cup \{j\}) - v(S_m)]$.
- **Preference-aware:** $\hat{\phi}_j^{\text{pref}} = (1/M) \sum_m [v_{\text{pref}}(S_m \cup \{j\}) - v_{\text{pref}}(S_m)]$.
 - Unbiased; variance = $\sigma^2/M \rightarrow$ MSE decays $O(1/M)$, absolute error $O(1/\sqrt{M})$.
 - **M = 50 selected: MSE $\approx 1.4 \times 10^{-5}$, ~99% accuracy on MovieLens-1M.**
 - Refreshed every 10 batches (~49 updates/epoch), smoothed by exponential moving average.

Objectives

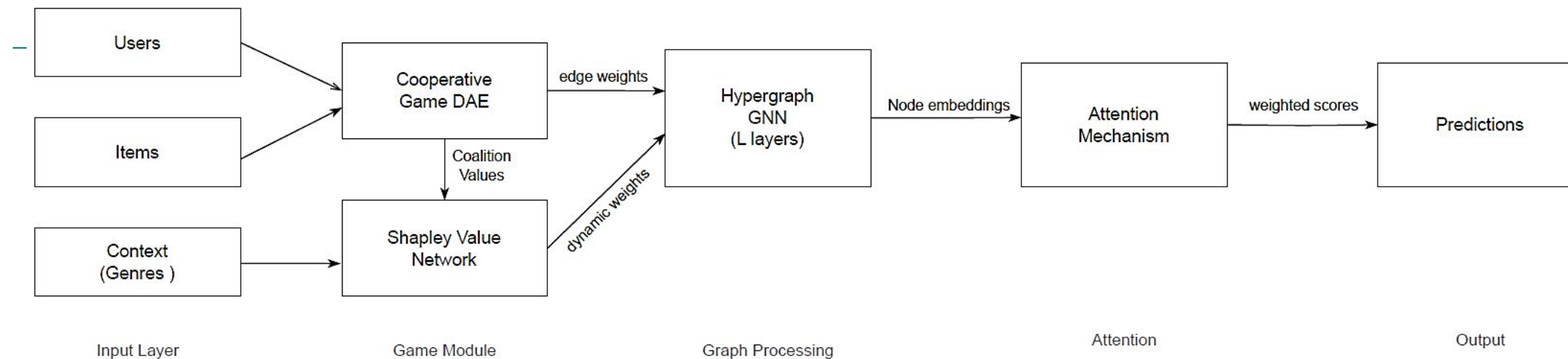
Methodology

Results

Findings

- Base propagation: $e^{(l+1)} = \sigma(D^{-1/2} A D^{-1/2} e^{(l)})$.
- **Shapley-weighted:** $e_j^{(l+1)} = \sigma(W^{(l)} e_j^{(l)} + \sum_{k \in N(j)} w_{jk} e_k^{(l)})$.
- **Normalised weights:** $w_{jk} = \hat{\phi}_{jk} / \sum_{k' \in N(j)} \hat{\phi}_{jk'}$.
 - Clipped + exponentially smoothed before normalisation (stabilises sparse regimes).

– Attention gate: $a_{ui} = \sigma(W_a [e_u || e_i || i])$; $v_{ui} = (1 + a_{ui}) / (e_u || e_i)$



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Objectives

Methodology

Results

Findings

- $L = L_{\text{rec}} + \lambda_{\text{div}} L_{\text{div}} + \lambda_{\text{ctx}} L_{\text{ctx}} + \lambda_{\text{reg}} L_{\text{reg}}$.
 - L_{rec} – Bayesian Personalised Ranking (pairwise, implicit feedback).
 - L_{div} – Intra-List Diversity regulariser: penalises redundant ranked lists.
 - L_{ctx} – Context alignment: match context embedding to context-node representation.
 - L_{reg} – L2 weight decay.
 - **The learning objective and coalition value are aligned: DyHuCoG trains to optimise the same balance that later determines attribution.**

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Contribution III — DyHuCoG: A Dynamic Hypergraph Cooperative Game

Objectives

Methodology

Results

Findings

Dataset	Model	NDCG@20	Recall@20	Coverage	Diversity
MovieLens-1M	HPCF	0.2528	0.2098	0.342	0.461
MovieLens-1M	DyHuCoG	0.2775	0.2362	0.397	0.516
Amazon-Book	HPCF	0.0270	0.0359	0.259	0.535
Amazon-Book	DyHuCoG	0.0306	0.0417	0.336	0.602

Metric	MovieLens-1M	Amazon-Book
NDCG@20	+9.77%	+13.33%
Recall@20	+12.58%	+16.16%
Coverage	+16.1%	+29.7%
Intra-List Diversity	+11.9%	+12.5%

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Objectives

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Findings

- **MovieLens-1M: Coverage 0.342 → 0.397 (+16.1%); ILD 0.461 → 0.516 (+11.9%).**
- **Amazon-Book: Coverage 0.259 → 0.336 (+29.7%); ILD 0.535 → 0.602 (+12.5%).**
 - Reduced filter-bubble effect and greater discovery opportunity — while NDCG/Recall also improve, so accuracy is not traded off for diversity.

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Contribution III — DyHuCoG: A Dynamic Hypergraph Cooperative Game

Objectives

Methodology

Results

Findings

Variant	ML-1M NDCG@20	% Drop	Amazon NDCG@20	% Drop
Full DyHuCoG	0.2775	–	0.0306	–
w/o Shapley Value	0.2647	4.6%	0.0287	6.1%
w/o Hypergraph	0.2586	6.8%	0.0279	8.9%
w/o Attention	0.2678	3.5%	0.0295	3.5%
w/o Context	0.2547	8.2%	0.0272	11.0%
w/o Diversity	0.2614	5.8%	0.0288	5.8%

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Objectives

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Findings

- Training: DyHuCoG ~2000 s vs HPCF ~1125 s on MovieLens-1M ($\approx 1.78\times$).
- Inference: 1.84 ms/query (ML-1M), 8.52 ms (Amazon) — suitable for real-time deployment.
- Memory: 4.4 vs 4.1 GB (ML-1M); 17.9 vs 16.8 GB (Amazon).
- Per-epoch cost: $O((L+1)md) + O((M/f)m)$.
- **Shapley convergence: $M=50 \rightarrow \text{MSE } 1.4 \times 10^{-5}$, ~99% accuracy; $M=100 \rightarrow \text{MSE } 3.5 \times 10^{-6}$ (diminishing returns).**

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Objectives

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Findings

- Paired t-tests on per-user NDCG@20 ($n = 6,040$ users; $df = 6,039$).
- DyHuCoG outperforms every baseline with extremely small p-values after Holm–Bonferroni correction.
 - **vs HPCF: $t = 46.38$, Cohen's $d_z = 1.3345$, $p = 1.81 \times 10^{-270}$.**
 - Wilcoxon signed-rank test also significant ($p < 0.001$).
 - Effect sizes are large — improvements are substantively meaningful, not merely statistically visible.

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Objectives

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Findings

- Cold-start (5 or fewer interactions): $\text{NDCG@20} \approx 0.061$ (user) and 0.057 (item), improving over HPCF by ~10%.
- Cross-dataset: MovieLens +9.9%, Amazon +14.8%, Yelp2018 +11.8%.
- **Interpretability: a SHAP waterfall decomposes a recommendation into ranking, diversity, context and preference contributions.**
 - Popularity bias: Shapley measures marginal utility, not raw frequency — weak but informative interactions retain influence.

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Objectives

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Results

Findings

▪ Answers to the questions and objectives

- RQ3 answered in the affirmative — attribution becomes an in-training signal, not a post-hoc diagnostic.
- RQ4 answered in the affirmative — ranking, coverage and diversity improve together.
- O1 met · recommendation formulated as a cooperative game over users, items and contexts.
- O2 met · preference-aware Monte Carlo Shapley embedded in message passing.
- O3 met · +9.9% (MovieLens) / +14.8% (Amazon) NDCG, with higher coverage and diversity.

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Objectives

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Findings

▪ Key findings

- Cooperative attribution used as an in-training signal — a stronger claim than the clustering chapters.
- The accuracy–diversity–context trade-off is not structurally fixed.
- NDCG, recall, coverage and diversity improve together on both datasets.
- Largest gains on the sparsest data (Amazon-Book), consistent with Shapley weighting helping when signal is weak.

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Objectives

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Findings

▪ Limitations

- Measurable computational overhead — roughly $1.78\times$ training time over HPCF.
- Depends on availability of meaningful context.
- Monte Carlo Shapley could be improved by variance reduction.
- Ablation is component-wise, so it does not test factorial interactions.
- Baselines finalised through early 2026, so superiority is claimed only against the tested baselines.

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Objectives

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▪ Takeaways

- Attribution is a first-class part of the learning objective, not a post-hoc diagnostic.
- The explanation is a direct read-out of the objective the model already optimises.
- **This makes the explanation structurally faithful rather than an external approximation.**
- Built on a principled, axiomatic basis — aligned with trustworthy-AI expectations.

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Conclusion & Perspectives

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Synthesis

Limitations

Perspectives

Conclusion

Contribution	Main idea	Achievement	Key finding
C1	Explain black-box clustering via Shapley	PCA–KMeans–LightGBM–TreeSHAP pipeline	Faithful, chemistry-consistent cluster attribution (wine)
C2	Multi-level, large-scale clustering XAI	Cross-level SHAP aggregation + Prop. 6.1	Coherent under hierarchy; interprets differences
C3	DyHuCoG hypergraph cooperative game	Preference-aware Shapley as in-training signal	Accuracy + coverage + diversity improve together

No.	Title	Venue	Status
I	Shapley Values for Explaining the Black Box Nature of ML Model Clustering	Procedia Computer Science 220, 806–811	Published, 2023
II	Game Theory Meets Explainable AI: An Enhanced Approach to Understanding Black Box Models Through Shapley Values	IJACSA 16(7), 716–725	Published, 2025
III	DyHuCoG: A Dynamic Hypergraph Cooperative Game for Preference-aware Recommendation	IJIES 19(2), 887–902	Published, 2026

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Conclusion & Perspectives

Synthesis

Limitations

Perspectives

Conclusion

- Computational – exact Shapley is intractable; every contribution relies on approximation, surrogates, or restricted reporting.
- Methodological – clustering depends on surrogate fidelity; recommendation depends on stable approximate contributions and adequate context.
- Empirical – tabular clustering + benchmark recommendation; no multimodal, sequential, or online deployment; no dedicated human-subject actionability study.
- **Claim scope – a coherent and productive perspective, not one fully unified framework eliminating all tension.**

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Conclusion & Perspectives

Synthesis

Limitations

Perspectives

Conclusion

- Scalable cooperative attribution – lower-variance Shapley, learned proposal distributions, adaptive refresh policies.
- Online / streaming recommendation – truly incremental settings with evolving graphs and delayed feedback.
- Richer human-centred evaluation – do explanations measurably improve analyst judgement, user trust, intervention quality, or perceived fairness?
- Broader trustworthy-AI evaluation – exposure fairness, transparency requirements, governance-oriented auditing.

- **Thesis answer: cooperative game theory can function as a shared methodological perspective for actionable explanation across clustering and recommendation.**

- Key outcomes:

- Shapley attribution as a common formal language for feature, interaction, and context importance allocation.
- Faithful clustering explanation, hierarchical explanatory coherence, and contribution-aware recommendation learning.
- Explanation as method, not commentary — from post-hoc description to in-training guidance.
- Aligned with trustworthy-AI requirements (EU AI Act, OECD principles, GDPR).

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Questions & Discussion

Thank you for your attention.

I welcome your questions and comments.